



TECHNO NJR INSTITUTE OF TECHNOLOGY

Name Harshita Chaplot Branch CSE Sem 4th
Roll No. 24ETCCS108 Date _____

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S.No.	Name of Program	Date	Signature
1.	Use of basic Unix shell commands: ls, mkdir, rmdir, cd, cat, touch		
2.	Write shell script for- (1) showing count of users logged in (2) Printing column list of files in home dir. (3) Listing job with below priority (4) Continue running your job after logging		
3.	Write shell script to change date format. Show time taken in execution of script		
4.	Write shell script to print file names in dir. showing date of creation & serial no.		
5.	Write shell script to count lines.		
6.	Write shell script to print end of Glossary		
7.	Write shell script to compute GCD & LCM of 2 numbers.		
8.	Write shell script to find whether given number is prime.		



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Roll No. _____ Date _____

Lab-1

Aim : Explain the following commands:
1. clear 2. cal 3. who 4. date
5. mkdir 6. ~~rm~~

Apparatus : Linux operating system

Standard Procedures :

Analyzing the Problem :

- 1) Start linux and enter the user and password.
- 2) Now write startx and after that open terminal.
- 3) At the terminal try different commands and see the output.

Designing the solution :

- 1) At terminal first perform the command CAL without and with different options available for it.
- 2) Like \$ cal and then enter. The calendar will displayed.
- 3) Same \$ cal -m then enter. In calendar Monday will displayed as first day of week.
- 4) Same way perform other commands like CLEAR, WHO, DATE, MKDIR, RM etc.

Implementing Solution :

1) CAL :

```
[user1@com]$ cal
```

```
[user1@com]$ cal -m
```



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```
[user@com]$ cat -j  
[user@com]$ cat -y
```

2) CLEAR :

```
[user@com]$ clear
```

3) WHO :

```
[user@com]$ who  
[user@com]$ who -g  
[user@com]$ who -H  
[user@com]$ who -m
```

4) DATE

```
[user@com]$ date  
[user@com]$ date -d '2 days ago'  
[user@com]$ date + %D  
[user@com]$ date + %d  
[user@com]$ date + %d %m %h
```

5.) MKDIR and RM

```
[user@com]$ cd Desktop/  
[user@com]$ ls  
[user@com]$ cd newfile/  
[user@com]$ ls  
[user@com]$ mkdir newfile1  
[user@com]$ ls  
[user@com]$ rm Sum-Of-Digits.txt  
[user@com]$ ls
```


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Name: Manjula Chakraborty Roll No. _____
Branch: _____ Date: _____
Sem: _____

Lab - 2

Aim :

Write shell script for -

- Showing the count of users logged in
- Printing column list of files in your home directory
- Listing your job with below normal priority
- Continue running your job after logging out.

Software used : Operating system - Linux

Source code :

i)

showing the count of users logged in,

wc -l | wc -l

echo "show all users logged in"

ii)

Printing column list of files in your home directory

ls -l | wc -l

iii)

Listing your job with below normal priority

ps -a

iv)

Continue running your job after logging out

nohup

command with options to note :-

nohup stands for no hang up

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Branch _____ Date _____
Sem _____

Lab - 3

Aim : Write a shell script to change data format, show time taken in execution of this script.

Software used : Operating system - Linux

Source code :

```
echo "Enter file name" read fname
echo "Input contains in $fname" cat > fname
echo "Display start file than current time" ls -l
    $fname
echo "Modification $fname" vi $fname
echo "Show start access time" ls -lt $fname
echo "Show modification time" ls -lt $fname
```



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Name Mahesh
Roll No. _____
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Lab - 4

- Aim : Write a shell script to print file names in directory showing dot of creation to serial no. of file.
- Software used : Operating system - Linux
- Source code :

```
echo "sort login name by time" &&  
echo "show login name"  
who  
echo "show only name and time" who | grep  
' $ print $ 1, $ 5 '  
echo "show sort by time" who | grep ' $ print  
$ 5, $ 1 '
```

• Output :

show only name and time root 12:45
show sort by time 12:45 root



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Name Prasanna

Roll No. _____

Branch _____

Date _____

Sem _____

Lab - 5

- Aim : Write a shell script to count lines, words & characters in its input.
- Software Used : Operating System - Linux
- Source Code :

```
read -p 'Enter file name ' fname
echo 'Input the contents of file ' cat > $fname
clear
echo 'Display all words ' cat $fname
echo 'Show file line, word, char ' gawk '
{ nc += length($0) + 1; nw += NF }
END { print "Line = " NF, "Word = " nw, "Char = ' nc ' ' $ fname
```


• Output :

Create and enter file name Display all records

11

22

33



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Kabo-6

- `vim` : write a shell script to print end of a glostary file in reverse order using `array`.
- software used : Operating system - linux
- Source code :

```
file_name = '$1'
# check if file name is given and exist if
[ "$#" -eq 0 ]
then
echo "Syntax : $0 filename" exit
elif [ 1 -f "$file_name" ] then
echo "File '$file_name' does not exist" exit
# Set the IFS variable to '\n' this enables
reading one in separated # line per word
IFS = $'\n' declare -a arr
# Read from file name and store each
line into next array location, while read
-h line do arr+=("$line"); done < "$file_name"
# If last line is not terminated read returns
false, body of while # is not executed. Instead of
adding it in the array directly print it. #
```



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because it will be the first line in the file.
Also make sure not to # terminate the line
with newline character, so use echo -n

if [! -z "\$line"] then
echo -n "\$line" fi

Count number of lines, and adjust index



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Name: Harish
Roll No. _____
Branch _____
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Sem _____

Lab-7

Aim :

Write a shell script to compute GCD & LCM of two numbers.

Software Used :

Source Code : Write a shell script to compute GCD of two numbers

echo "Enter two numbers with space in bet read a b"

m = \$a

if [\$b -lt \$m] then

m = \$b fi

while [\$m -ne 0] do

x = 'expr \$a % \$m' y = 'expr \$b % \$m'

if [\$x -eq 0 -a \$y -eq 0] then

echoed "GCD of \$a and \$b is \$m break

fi

m = 'expr \$m - 1' done

Write a shell script to compute LCM of 2 numbers

echo "Enter first no" read a

echo "Enter 2nd no" read b

p = 'expr \$a % \$b' while [\$b -ne 0] do

a = 'expr \$a % \$b' a = \$b

b = \$p done

LCM = 'expr \$p / \$a' echo "LCM = \$LCM"

Lab-8

Aim :

Write a shell script to find whether a given number is prime.

Software Used : Operating System - Linux

Source Code :

```
i=2
num=1
echo "Enter a number" read num
if [ $num -lt 2 ] then
echo -e "\$num is not prime\n" exit 0 fi
while [ $i -le `expr $num / 2` -a $num -ne 0 ]
do
num=`expr $num % $i`
if [ $num -ne 0 ] then
echo -e "\$num is prime\n" else
echo -e "\$num is not prime\n" fi
i=`expr $i + 1`
done
```